

NextBillion AI: Solving hyperlocal AI needs around the world

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NextBillion AI improves time to market for hyperlocal AI solutions by running datasets and algorithms on Cloud Storage and Cloud Storage and Cloud minimizes operational overheads with Google Kubernetes Engine



About NextBillion AI

NextBillion AI is an artificial intelligence (AI) platform that helps customers build AI-powered hyperlocal applications, without investing in infrastructure. It offers two solutions, NextBillion Maps to build location-based experiences based on historical and streaming data and NextBillion Tasks to carry out tasks such as multilingual text annotation with AI and human insights.

Nextbillion AI wants to empower internet users with equitable and affordable access to technology.

Industries: Technology

**Google
Cloud
results**

**Enable
and-pla**

- Improves speed to market for features releases with CI/CD pipelines on Google Kubernetes Engine
- Delivers 99.95% uptime on highly available Google Cloud to minimize service disruption to customers
- Simplifies on-demand cluster creation with Dataproc to accelerate compute-intensive AI workloads

to meet custom specific

Location: Singapore

(<https://cloud.google.com/sql/>), [Cloud](#)



About Search

Search is a Google Cloud Premier Partner on a mission to futurify businesses by leveraging cloud, automation, and analytics. It helps businesses through digital transformation by replacing on-premises applications with cloud solutions.

Google Cloud (<https://cloud.google.com/>)

Cloud IAM (<https://cloud.google.com/iam/>)

Cloud SQL (<https://cloud.google.com/sql/>)

Cloud Storage (<https://cloud.google.com/storage/>)

Cloud CDN (<https://cloud.google.com/cdn/>)

Kubernetes Engine
(<https://cloud.google.com/kubernetes-engine/>)

Cloud Dataproc
(<https://cloud.google.com/dataproc/>)

Operations
(<https://cloud.google.com/products/operations>)

Logging (<https://cloud.google.com/logging/>)

The internet can improve people's lives by providing access to information and digital opportunities.

According to Bain, between 2013 to 2017, nearly one billion people used the internet for the first time. By 2025, another billion users

(<https://media.bain.com/next-billion-internet-users/index.html>)

are expected to come online from emerging markets in Asia, Africa, and the Middle East.

Even though they may be equipped with smartphones, customers in remote areas may be isolated geographically. The lack of local language content can isolate them further. Instead of offering a homogeneous mobile app developed for urban cities, companies can attract rural customers with a location-based experience that adapts to their local needs, from last-mile delivery to native language support.

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*—Gaurav Bubna, co-
founder, NextBillion AI*

NextBillion AI (<https://nextbillion.ai/>) helps companies connect to users with hyperlocal AI solutions.

NextBillion AI Maps offer custom map solutions through a range of APIs such as routing and navigation that integrate with the customer’s mobile or web app. NextBillion AI Tasks combines artificial intelligence (AI) and human intelligence to carry out tasks such as decoding and simplifying multilingual text, classifying images, and annotating videos.

“Building an AI solution is a big commitment, from recruiting AI talent to investing in the infrastructure. Some companies may develop AI in-house for one or two critical business functions, but it’s hard to invest at the same level for non-core tasks,” says Gaurav Bubna, co-founder at NextBillion AI. “That’s

where we add value as a third party with the people, process, and AI capabilities on Google Cloud to take on non-core tasks such as mapping in a cost-effective manner.”

“We built a nimble AI platform on Google Cloud (<https://cloud.google.com/>) that allows clients to choose plug and play modules, depending on their geography and use case,” says Ajay Bulusu, co-founder at NextBillion AI. He explains that while some clients may want to detect road names and traffic signs from street-level imagery to improve delivery accuracy, others may want to annotate a politician’s name to uncover public sentiment from news articles and public forums.

NextBillion AI deploys its data pipelines on Google Kubernetes Engine (/kubernetes-engine) to ensure high uptime for clients and minimize maintenance with auto updates. The company stores client datasets in Cloud Storage (/storage) and uses Cloud SQL (/sql) to run real-time queries.

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*—Chang Zhao, Head of
Engineering, NextBillion
AI*

Simplifying DevOps processes with Google Kubernetes Engine and Dataproc

NextBillion AI started with self-hosted Kubernetes in 2019, but as the number of projects increased, the lean DevOps team found it challenging to maintain the infrastructure and run concurrent pipelines. The company decided to migrate to a managed

Kubernetes solution to keep up with business growth.

With the help of Google Cloud partner Searce (<https://www.searce.com/>), NextBillion AI ran a two-week proof of concept to evaluate cloud service providers based on feature richness, support effectiveness, and its ability to follow the latest updates. Now, Google Kubernetes Engine (GKE) provides native support for Kubernetes, keeping clusters up to date with the latest versions.

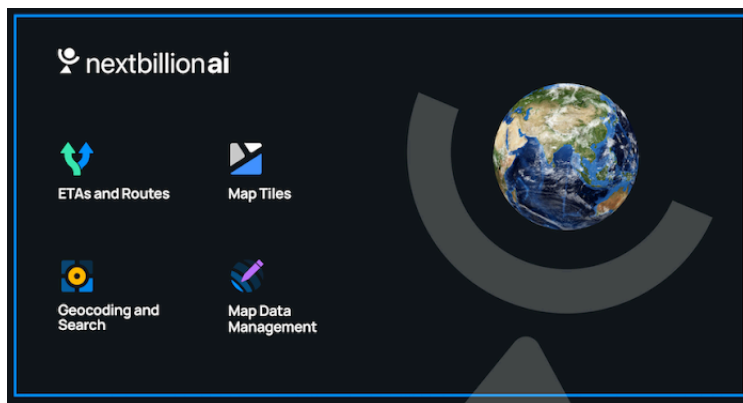
At 99.95% uptime, NextBillion AI can only allocate a 20-minute maintenance window each month to roll out features and updates. Chang Zhao, Head of Engineering at NextBillion AI, explains, “Tasks such as scaling up and down, rollouts, and rollbacks used to require a lot of scripting work on our legacy platform. Google Kubernetes Engine simplifies DevOps processes such as auto-scaling and rolling updates, so we don’t need a large team to maintain the infrastructure and pipelines.”

The company also speeds up new feature releases such as Route Optimization API by automating continuous integration and delivery (CI/CD) pipelines on Google Kubernetes Engine. Once the developer changes the code, an automated test will run so bugs can be identified and fixed before the feature goes live. With continuous delivery (</solutions/continuous-delivery>), NextBillion AI can roll out updates to minimize planned downtime for clients and their end customers.

As a service provider, NextBillion AI pays close attention to its business service-level agreement (SLA) commitments such as response and resolution time. It uses Cloud Logging (/logging) to store logs from VM instances such as output error messages that can be viewed in real time so the admin can do a deep dive into online issues and resolve them quickly.

Data scientists at NextBillion AI also use Datapro (/dataproc) to manage compute-intensive AI workloads, without the help of DevOps to set up the environment. From ingesting to training data, Datapro makes on-demand cluster creation a breeze. Using programmer-friendly command lines, data scientists can start, scale, or shut down a cluster in seconds on Datapro merely by running a script. When processing is complete, the data scientist moves the data to Cloud Storage and shuts down the whole cluster without incurring extra costs on idle clusters.

“Datapro matches our business needs. We’re not running one job that streams petabytes of data into a cluster at all times. Instead, we run multiple AI pipelines to process data or train models for different customers in parallel,” says Chang.



Running AI pipelines at scale

A significant challenge of deploying AI workloads is the massive volume of data. For example, to predict traffic conditions for a logistics customer, NextBillion AI needs to run algorithms on historical traffic patterns and live traffic conditions. The company stores hundreds of AI algorithms, ranging from millions to billions of rows, on Cloud Storage for big data processing such as extract, transform, and load (ETL) process, AI and machine learning (ML) training.

As part of the onboarding process, NextBillion AI gives clients access to the portal containing dedicated folders where they can upload raw data to Cloud Storage for ETL. The next step is to extract necessary features for ML training and model training. The AI company returns the processed data back to the customer through APIs for NextBillion AI Maps or CSV files for NextBillion AI Tasks.

Chang says, “AI datasets are typically too large to share on a thumb drive or a spreadsheet. We set up Cloud CDN (/cdn) on Cloud Storage to speed up the flow of data for our pipelines so that clients can

upload multiple terabytes of data, or even petabytes of data, from their locality across regions to secure Cloud Storage buckets efficiently.”

Balancing security and privacy with convenience

One question that prospective clients often ask NextBillion AI is, “Will you use my data to power maps for other businesses?” The AI company addresses data protection concerns by providing all clients with Cloud Identity and Access Management (/iam).

“We run a secure pipeline in Google Cloud. Clients have fine-grained control over who has what access to their folder or bucket in Cloud Storage,” says Ajay. “Client data is kept in different folders so no action by one client can affect another client.”

The companies that NextBillion AI works with need to comply with strict data protection rules such as General Data Protection Regulation (GDPR) because they collect personal data. Such companies can only work with GDPR-compliant third parties to ensure consistency in how personal data is collected, stored, and managed. NextBillion AI helps companies meet compliance requirements by recommending best practices such as removing personally identifiable information (PII) from the raw data and by processing data on Google Cloud.

“Today, no machine can process data at such scale and run complex AI algorithms at speed on a single node. Our distributed computing architecture on Google Cloud gives us a competitive advantage and positions us well for the future.”

—Ajay Bulusu, co-founder,
NextBillion AI

Improving AI solutions for large-scale projects

As next steps, the company is exploring Google Cloud AI and ML tools such as [Vision AI](#) (/vision) for

recognizing images as it moves into new business areas such as content and facial recognition. The plan is to help marketers overcome content-related challenges such as security and monetization with the power of AI. It will use facial recognition AI technology to address identity verification problems in FinTech, keeping in mind emerging use cases such as face mask detection and fever detection.

“Google Cloud helps us scale effortlessly as the number of projects increases. We expected to see individual file sizes in the petabyte range as geospatial and content use cases expand,” says Ajay. “Today, no machine can process data at such scale and run complex AI algorithms at speed on a single node. Our distributed computing architecture on Google Cloud gives us a competitive advantage and positions us well for the future.”

